

TS-Cast-NIO v2

The shipped model: a 3-D satellite encoder compressing eleven days into 128 numbers, and a compact head turning those into fifteen depths with an uncertainty on each.

PORT	PAGE FILE	MODEL	INPUT
8507	app/phase2/tscast_page.py	548,582 parameters	7 satellite channels, 11 days

This is the feature the problem statement actually asked for, and everything else in this pack either feeds it, checks it or uses its output. It reconstructs fifteen depths of temperature across the entire North Indian Ocean, every day, from satellite observations alone.

1. What this feature is, in plain words

The model sees a stack of seven satellite pictures - sea-surface temperature, salinity, height, two current components and two wind components - for eleven consecutive days around the date you ask about. A 3-D convolutional encoder compresses that stack into 128 numbers: a compact description of what the surface is doing there.

Those 128 numbers are the **compact satellite embedding** the problem statement asks for, and they are the whole of what the network knows about that place and day. A small head then turns them into fifteen temperatures and fifteen uncertainties. Nothing else enters: no climatology, no reanalysis, no in-situ float.

That last point is a measured departure from the paper we re-implemented, and it is worth stating plainly.

TS-Cast's design feeds the monthly climatology into the decoder so the network only has to learn a correction to a physically sensible average. We built that decoder, scored it, and it was **worse** at our data scale - so it is not shipped, and the shipped network predicts the profile directly from the satellite latent. Verified by running the model with the climatology set to zero and to noise and getting bit-identical output.

Alongside every temperature it emits a second number: how uncertain it is at that depth. That is why Feature 13 exists at all - the uncertainty is produced by the model itself, not bolted on afterwards.

It is a re-implementation from the published description of TS-Cast (Chae, Donohue and Park, *Ocean Science* 22, 2026). Their code was never released; nothing is copied. And we depart from the paper in several places - each departure measured, not assumed.

THE EVERYDAY PICTURE

Squeeze eleven days of satellite imagery over a patch of sea into 128 numbers - about the size of a short sentence - and that sentence has to contain everything needed to describe the kilometre of water underneath. The encoder writes the sentence; the head reads it back out as a temperature profile.

2. What you actually see on the screen

The model's own dashboard, with the ground-truth check beside every number:

- Reconstructed temperature at any depth across the basin, for any date in the bundle.
- The per-depth error table: RMSE, correlation and bias against 962 independent Argo profiles.
- The skill-against-climatology chart, with **both** skill definitions always shown and always labelled, because they read very differently and mixing them would be misleading.

- The provenance block - which checkpoint, which data bundle, which commit, and the checkpoint's own cryptographic hash.
- An explicit **input_source** field that reads **satellite**, which is the field the freeze script asserts before it will accept the model as the deliverable.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/tscast_page.py --server.port 8507
```

- 1 Run the command and open **http://localhost:8507**.
- 2 Read the provenance block first, out loud. It names the exact checkpoint and says **input_source: satellite**. This is the moment you prove the deliverable is what it claims to be.
- 3 Pick a date and a depth, and let the basin render.
- 4 Open the per-depth error table and walk down it. Point at 100 m - the hardest depth - and at 200 m, where the model gains most over climatology.
- 5 Show the skill chart with both definitions side by side and explain why you never quote one beside the other.
- 6 To prove the model file has not drifted, run the freeze check in a second terminal.

VERIFY THE SHIPPED MODEL - 18 CHECKS

```
PYTHONPATH=src python scripts/phase2/freeze.py --check
```

RE-SCORE THE CHECKPOINT FROM DISK, TRUSTING NO SIBLING FILE

```
PYTHONPATH=src python scripts/phase2/rescore_checkpoint.py --checkpoint artifacts/tscast_stage1.pt
```

RETRAIN IT - the --daily-dir flag is NOT optional

```
PYTHONPATH=src python -m phase2.tscast_nio.train.train_stage1 \
  --daily-dir data/processed/daily_sat/v001 --t-seq 11 --epochs 25 \
  --train-samples 60000 --test-samples 12000 --patience 5 \
  --encoder cnn3d --decoder simple --tag sat_7ch_s42
```

WHY --DAILY-DIR IS NOT OPTIONAL

Omitting it silently trains on the reanalysis instead of on satellite data, which would produce a better-looking number that does not satisfy the problem statement. The flag is required, and the freeze script asserts `input_source == satellite` on the shipped artifact so this cannot regress unnoticed.

4. Why the problem statement needs this

It is the requirement. The problem statement asks for a compact satellite embedding produced by deep learning, feeding a reconstruction to fifteen standard depths at 0.25 degrees, daily, over the North Indian Ocean, evaluated by RMSE, correlation and bias against independent observations. This feature is all of that in one artifact.

There is also a decision here that a jury should hear explicitly, because it is where scientific integrity cost us accuracy. Our **most accurate** configuration scores 0.8548 degC - but it is fed reanalysis inputs, and the problem statement asks for satellite observations. We do not ship it and we do not quote it as our result. The shipped model scores 0.9006 degC on genuine satellite input. Against the reanalysis-fed stage-1 comparator, real observations read +0.026, +0.027 and -0.003 degC on seeds 42, 43 and 44 - a mean of +0.017 degC whose **sign does not hold**, so by our own three-seed rule the cost is not an established effect: the two inputs are within seed noise of each other, and the satellite model retains about 94% of the comparator's skill.

PS requirement	How this feature answers it
Compact satellite embedding via deep learning	A 3-D CNN encoder producing a 128-dimensional latent, chosen over four competing architectures in a controlled bake-off.
Reconstruct surface -> full profile	One forward pass returns all 15 depths for every grid cell.
0.25 degrees, daily, 5-30 N / 45-105 E	100 x 240 grid, 388 consecutive daily steps, zero gaps.
RMSE, correlation and bias	All three at all 15 of 15 depths against 962 independent Argo profiles.
Training target: GLORYS reanalysis	GLORYS12V1 daily, with the target's own error separately measured in Feature 3.

5. The measured numbers behind it

The shipped result, on 962 independent Argo profiles never used in training, 12,736 depth comparisons [VERIFIED, scoring protocol seafloor_masked_v1]. The 962 are the profiles within 5 days of the test window; 908 fall strictly inside it, and scoring those alone reads 0.8978 degC. The scorer declines the 93 comparisons (0.7%) at depths where the training target has no water and the product itself returns nothing, plus one profile on a land cell; scored WITH them, as every artifact before 2026-09-07 was, the same checkpoint read 0.9078 degC and skill +0.2595:

Metric	Value	How to read it
RMSE	0.9006 degC	The headline. Average error across all depths.
Correlation	0.8809	Mean of the 15 per-depth values.
Bias	+0.1066 degC	Positive means the model runs warm. See Feature 15.
Skill vs climatology	+0.2400	1 - RMSE/RMSE_clim, where RMSE_clim = 1.1850. Where the cell has a REAL per-cell climatology (895 of 962 profiles) it is +0.1494 ; the other 67 sit on a basin-mean fill, and beating a fill is not skill.
Murphy skill	+0.4225	1 - MSE/MSE_clim. Never quote one beside the other.
Depths beaten climatology	14 of 15	At 1000 m climatology wins by 0.055 degC, and the chart says so.

- **The error has a shape, and the shape is the story.** It is small at the surface (0.47 degC at 5 m), bulges through the thermocline (peak **1.22 degC at 100 m**, where a surface field constrains depth least), and collapses below 500 m (0.31 degC at 1000 m). The widest gain over climatology is **+0.53 degC at 5 m**.
- **Two measured disagreements with the paper we re-implemented, and one withdrawn.** Its density constraint costs accuracy (0.8593 against 0.8548 with it off), and its FiLM decoder was measured to cost accuracy, so it is built, tested and deliberately **not shipped** - the code refuses to run stage 2 on it with an explicit error message saying why. Both rest on **one seed each**, which by this project's own three-seed rule is suggestive rather than settled. **The input-window comparison is WITHDRAWN**: all three legs predate our leakage fix, the 31-day leg is on our own do-not-quote list, and the two shorter legs' checkpoints were overwritten. We ship an 11-day window and cannot presently prove it is the best one.
- **The model is deliberately small.** At the paper's widths it would be 7,118,474 parameters - 13.1x the capacity for our 100,000 samples - and it overfits by epoch 3 regardless of the loss function. We ship 548,582 parameters.

6. The impact - and who it is for

Who	What changes for them
INCOIS / MoES	Daily basin-wide subsurface temperature from satellites already in operational use - no new instrument, no new deployment.
Cyclone forecasting	The subsurface heat field that drives intensification, available the same day rather than after a float surfaces.
Fisheries and shipping	Thermocline and layer structure everywhere, every day, instead of only where a float happens to be.
Ocean research	A reproducible, frozen, independently-validated reconstruction over a basin where such products are scarce.

WHY SOCIETY GAINS FROM THIS

Argo floats measure this basin roughly every five to ten days, and only where a float happens to be. This produces the same quantity at 24,000 locations every single day from satellites that already exist. For a coastline where tens of millions of people live inside a cyclone's reach, the difference between 'somewhere, sometimes' and 'everywhere, daily' is the entire point of the project.

7. Say this to the jury

This is the deliverable. Seven satellite channels over eleven days go in; fifteen depths of temperature across the whole basin come out, with an uncertainty on every one. Against 962 Argo floats it never saw, it scores 0.9006 degC and beats climatology at fourteen of fifteen depths. And I want to be explicit about one choice: our most accurate model scores 0.8548, but it is fed reanalysis, and the problem statement asks for satellite observations. We do not ship it and we do not quote it. Across three seeds, real observations read between -0.003 and +0.027 degC against it - a mean of +0.017 whose sign does not hold, so we do not claim a cost either: the two are within seed noise.

The refusal to quote 0.8548 is the single most credible thing you can say.

8. Questions the jury will ask

Q. Why is your headline number not your best number?

A. Because our best number is fed reanalysis input and the problem statement asks for satellite observations. Quoting it would present a reanalysis-fed model as satisfying a satellite requirement. Our freeze script asserts `input_source == satellite` on the shipped artifact, so this cannot regress silently even if someone wanted it to.

Q. The paper anchors its decoder on the climatology. Why does yours not?

A. Because we measured it and it was worse at our data scale. We built the paper's climatology-prior decoder, scored it against Argo, and it cost accuracy - so we ship the head that was actually validated. It is one of three measured disagreements with that paper, and we can show the evidence for each. A useful side effect: because no climatology enters the shipped network, the sensitivity analysis in our observability work measures the satellite dependence and nothing else.

Q. You re-implemented a 2026 paper. What is your own contribution?

A. Three things, and we are careful not to overclaim. First, the region: the model, the climatology and the validation are all built for the North Indian Ocean. Second, the measured disagreements - the density constraint and the FiLM decoder both hurt at our data scale, on one seed each, and the input-window comparison is withdrawn because it predates our own leakage fix. Third, the system around it. We claim a system-level contribution, not a new reconstruction method, and our own novelty matrix says so in writing.

Q. How do we know the checkpoint on screen is the one you scored?

A. Run `freeze.py --check`. It re-verifies eighteen properties of the shipped artifact rather than trusting the manifest, including the checkpoint's SHA-256 hash and the claim wording used for uncertainty.

9. The honest limit - say it before they find it

WHERE IT IS WEAKEST, IN ITS OWN WORDS

It runs **warm** - bias +0.1003 degC overall, peaking at +0.657 degC at 50 m - but that average is **inherited**: the GLORYS target is +0.1078 degC warm against the same floats, and the model is -0.007 against its own target. What is ours is the shape, and at 50 m the model adds **+0.447 degC** on top of the target's +0.213. Its mixed layer (20-50 m) is genuinely worse than the reanalysis by **0.23 to 0.38 degC**. At the thermocline **most** of the error is inherited - the reanalysis itself scores 1.042 degC at 100 m against our 1.218 - but **0.178 degC of it is ours**, not the 0.023 an earlier draft carried over from a Phase-1 measurement. And there is an Arabian Sea satellite penalty of +0.0341 degC whose sign holds across all three seeds and whose **cause is UNKNOWN after four tested hypotheses** - we report it as unexplained rather than offering a story.

IN ONE LINE

The deliverable: seven satellite channels over eleven days in, fifteen depths of temperature with uncertainty out, straight from a 128-number satellite embedding - and the more accurate reanalysis-fed model deliberately not shipped.

0.9006 degC

RMSE VS 962 INDEPENDENT PROFILES

+0.2400

SKILL OVER CLIMATOLOGY (+0.1494 WHERE
THE BASELINE IS REAL)

548,582

PARAMETERS - 13X SMALLER THAN THE
PAPER'S

Feature 7 of 17. Next: Feature 8.